

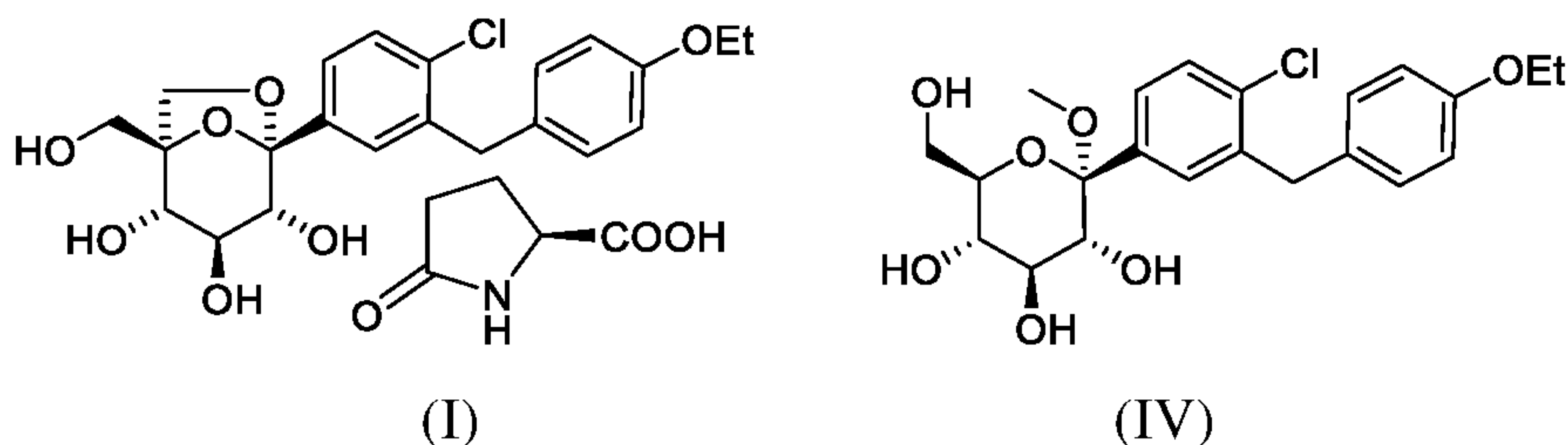
“AN EFFICIENT PROCESS FOR THE PREPARATION OF ERTUGLIFLOZIN L-PYROGLUTAMIC ACID AND INTERMEDIATES THEREOF”

RELATED APPLICATION

This application claims the benefit of Indian Provisional Application No. IN201921054634, filed on December 31, 2019, the contents of which are incorporated by reference herein.

FIELD OF THE INVENTION

The present invention relates to an efficient process for the preparation of Ertugliflozin L-pyroglutamic acid of formula (I) and its intermediates in environment friendly manner. The present invention further relates to a process for the preparation of substantially pure intermediate of formula (IV).



BACKGROUND OF THE INVENTION

Ertugliflozin, is chemically known as (1S,2S,3S,4R,5S)-5-{4-chloro-3-[(4-ethoxyphenyl)methyl]phenyl}-1-(hydroxymethyl)-6,8-dioxabicyclo[3.2.1]octane-2,3,4-triol. Ertugliflozin is a sodium-glucose co-transporter-2 (SGLT2) inhibitor approved in the US on Dec 19, 2017 to improve glycemic control in adults with Type-2 diabetes. Ertugliflozin and its co-crystal Ertugliflozin L-pyroglutamic acid (Ertugliflozin LPGA), Ertugliflozin L-proline are disclosed in the US Patent No. 8,080,580 (henceforth '580).

US '580 and US patent No. 8,669,380 (henceforth '380) discloses the preparation of Ertugliflozin, its analogues and Co-crystals thereof [Scheme-2], generically US